COOLING ENVIRONMENTS

Considerations for Shipping and Storage of 1-Phase Cold Plate Liquid Cooled Racks and Servers

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Introduction

Why - Liquid Cooled hardware needs special attention on shipping and storage process.

What – this presentation includes

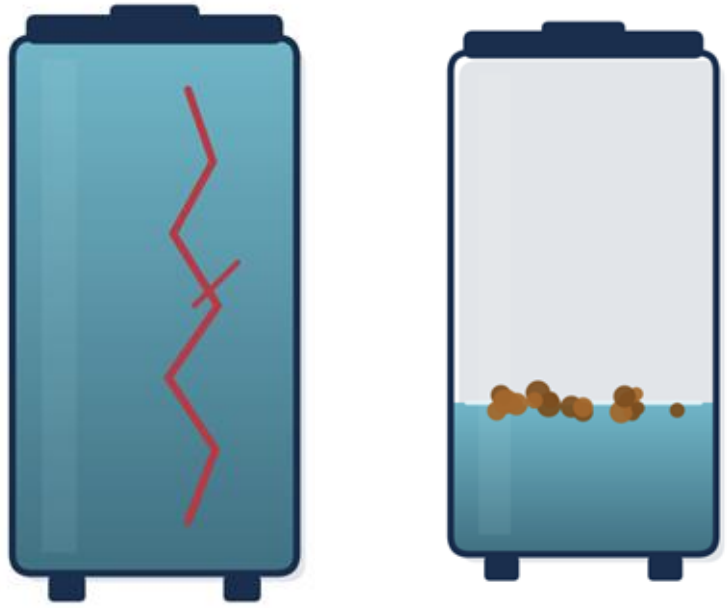
- Shipping and Storage Options
- Technical Risks
- Operational Solutions
- At Scale Considerations

Goal – alignment on hardware design and operation process, to tackle the logistic complexities

Why

Technical Challenges

- Coolant expansion risk
- Corrosion risk
- Coolant health management

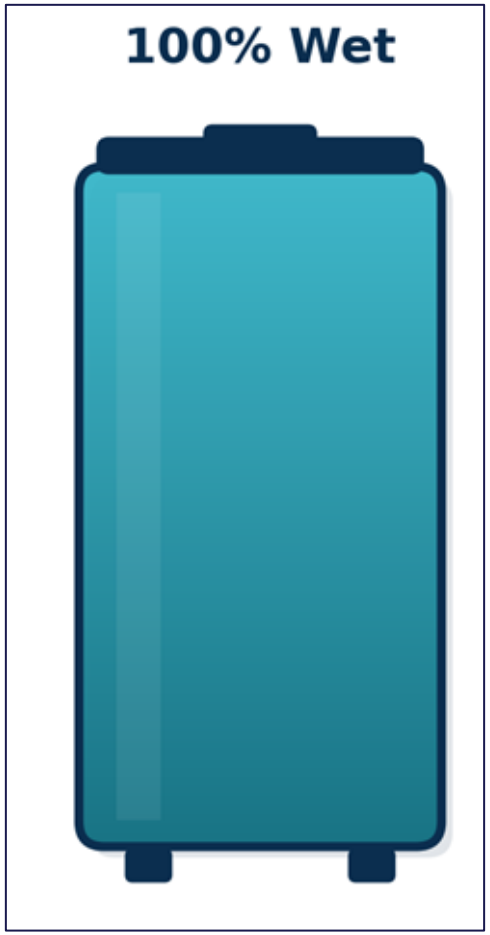


Four Options:

Different modes based on liquid ratio and treatment process



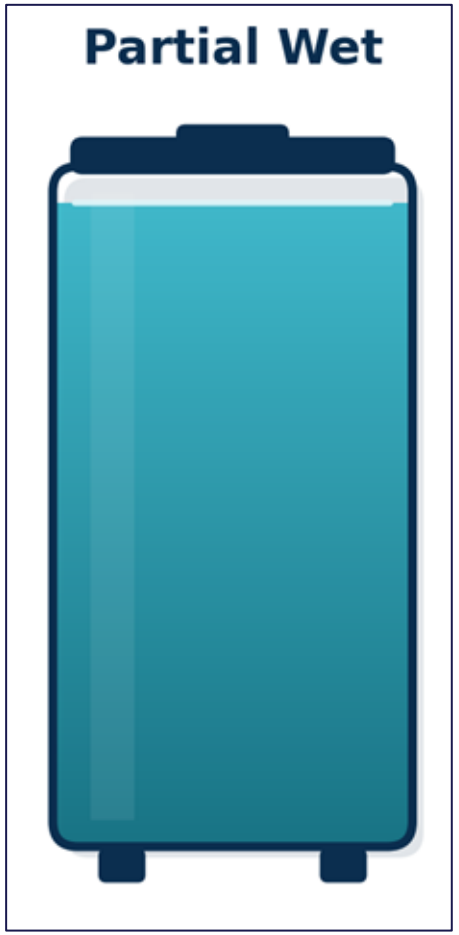
Four Options:



Wet Mode – ready upon arrival

- Fully filled with coolant
- Faster power on without prefill
- Volume expansion risk apply; temperature control needed

Four Options:



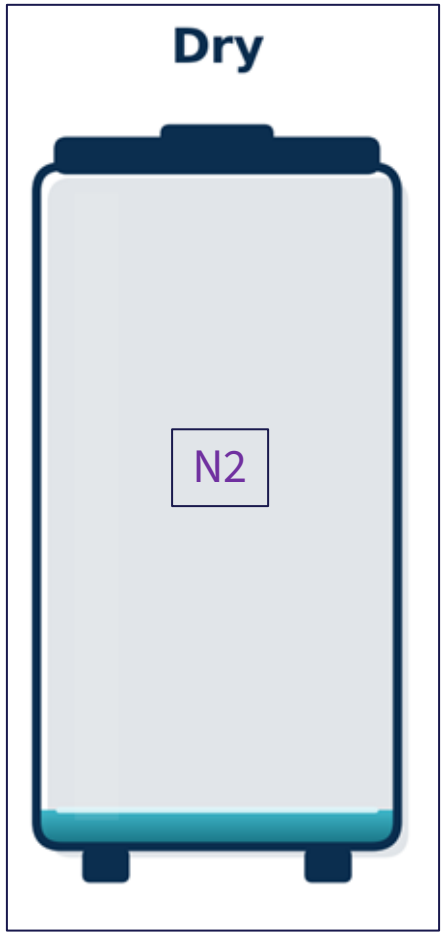
Partial Wet Mode – top off needed

- 90%~95% filled with coolant, rest being air
- Top off needed before power on
- Corrosion risk applies; constrained shelf life
- Temperature control still needed

Four Options:

Dry Mode – operational heavy

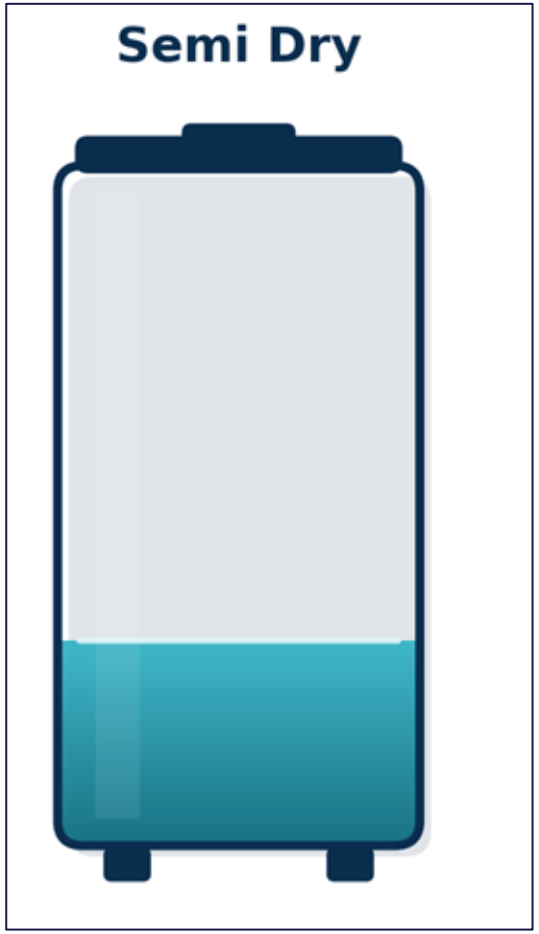
- Properly drained and filled with N2
- N2 release and coolant prefill needed before power on
- No need of temperature control
- Corrosion risk is contained



Four Options:

Semi Dry Mode – quick turn around

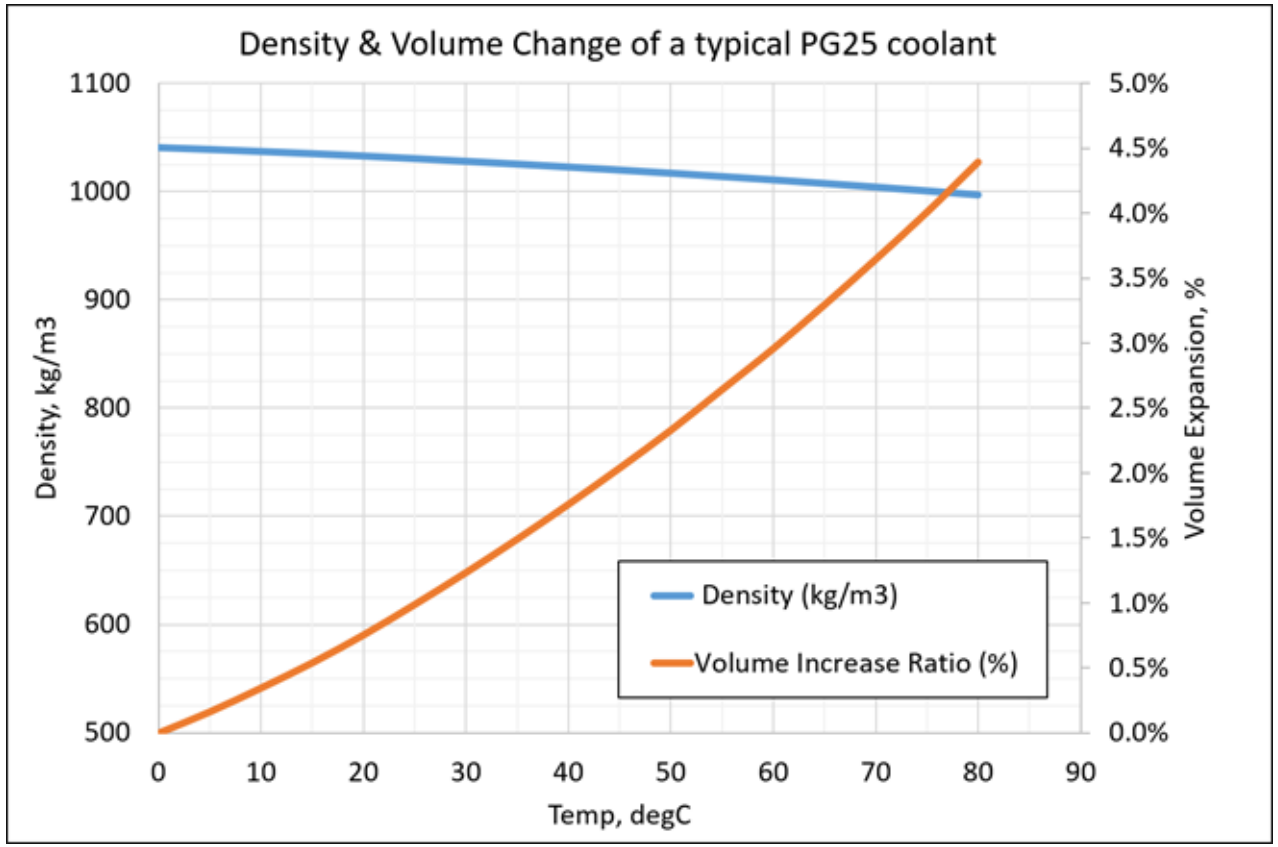
- Applicable to sites without nitrogen filling station
- Drain with compressed air before shipping
- No need of temperature control
- Corrosion risk is high, very limited shelf life



Trade Off Summary

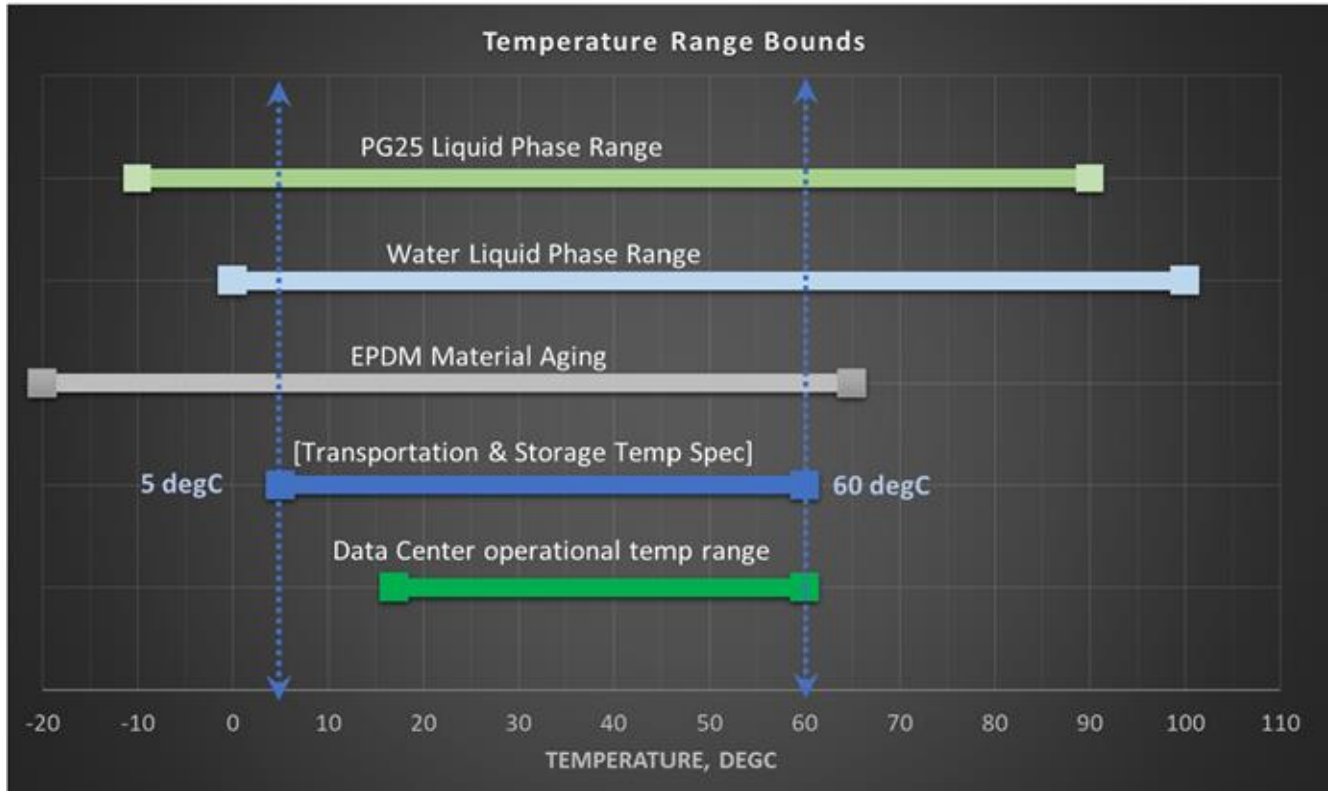
Shipment	Wet	Partial Wet	Dry	Semi Dry
Liquid Fill Ratio	**Almost 100%**	**90%~95%**	**< 10%**	**< 50%**
User Scenario	General Inbound or Outbound	Early NPI Inbound	General Inbound or Outbound	Fast Outbound RMA
Technical Risk				
Risk of Expansion Damage	Medium	Low	Low	Low
Risk of Corrosion	Low	Medium	Low	High
Operational Burden				
Prepare - Drain Liquid	N	Y	Y	Y
Prepare - Charge N2	N	N	Y	N
Shipment - Temp Control	Y	Y	N	N
Before Use - Topoff/Prefill	N	Y	Y	Y
Storage	**Wet**	**Partial Wet**	**Dry**	**Semi Dry**
Technical Risk				
Risk of Expansion	Medium	Low	Low	Low
Risk of Corrosion	Low	High	Medium	High
Operational Burden				
Warehouse -Temp Control	Y	Y	N	N
Warehouse-Shelf Life	Long	Short	Medium	Short
Warehouse - N2 Renew	N	NA	Y	NA
Before Use - Prefill	N	Y	Y	Y

Fluid Expansion



- As temperature increases to boiling points, both PG25 and water expand approx. 4%
- As temperature decreases to freezing points
  - PG25 expands approx. 2-5%
  - Water expands approx. 9%
- Freeze expansion is more risky than high temperature expansion in practice
- Validation against transit thermal variations is required for wet shipment.

Temperature Control



Temperature Control is critical for transportation and storage of ‘wet’ hardware

- Non-operational temp range based on coolant phase limits and typical data center conditions.
- Packaging thermal inertia provides buffer against short environmental excursions.

Different approaches to transportation

- Reference standard: IEC 60721-3-2
- Engage shipping provider to confirm temperature equipment will be exposed to.
- Instrument a dummy rack with thermal sensors during freight shipment to verify thermal variations.

Shipment Method	Temperature Range (Standard)	Temperature Range (Controlled)
Air Freight	-10°C (14°F) to 40°C (104°F)	5°C (41°F) to 25°C (77°F)
Road Freight (Truck)	-20°C (-4°F) to 45°C (113°F)	5°C (41°F) to 25°C (77°F)
Rail Freight	-20°C (-4°F) to 45°C (113°F)	5°C (41°F) to 25°C (77°F)
Sea Freight	-10°C (14°F) to 50°C (122°F)	5°C (41°F) to 25°C (77°F)

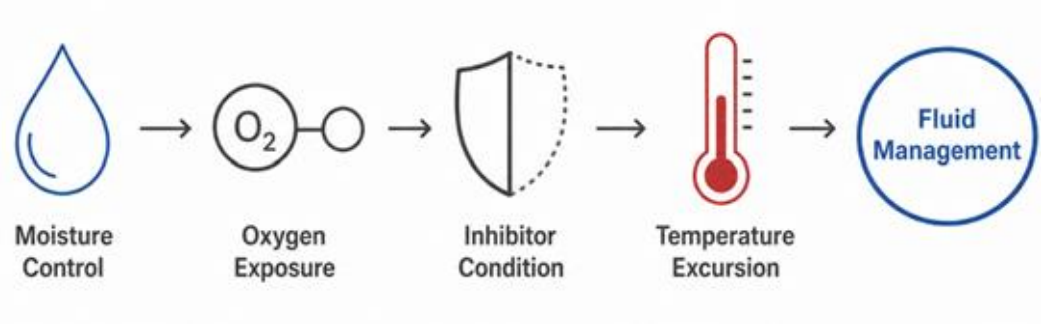
Fluid Management

Ship Wet

- Inhibited coolant provides robust corrosion protection
- Established passivation on wetted surfaces
- Control storage temp (20–25°C) for extended durations

Ship Dry

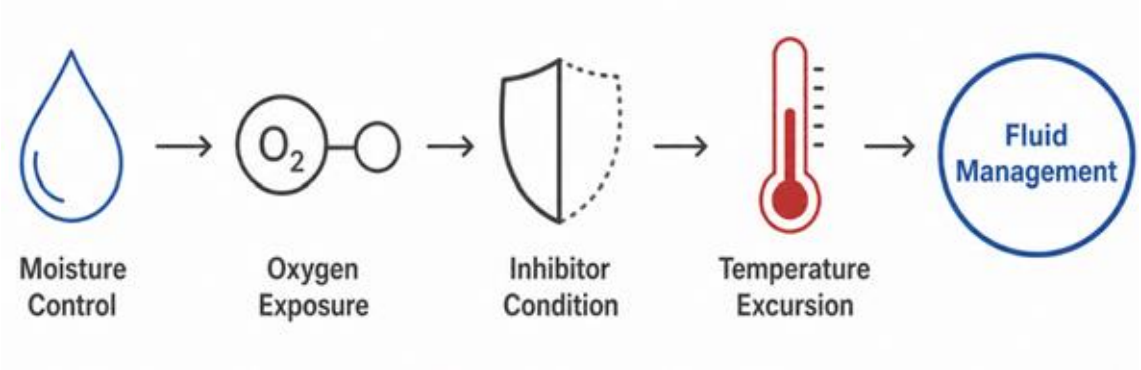
- Inert atmosphere protects internal surfaces
- Pre-passivation prior to nitrogen purge
- Onsite fluid fill required at deployment



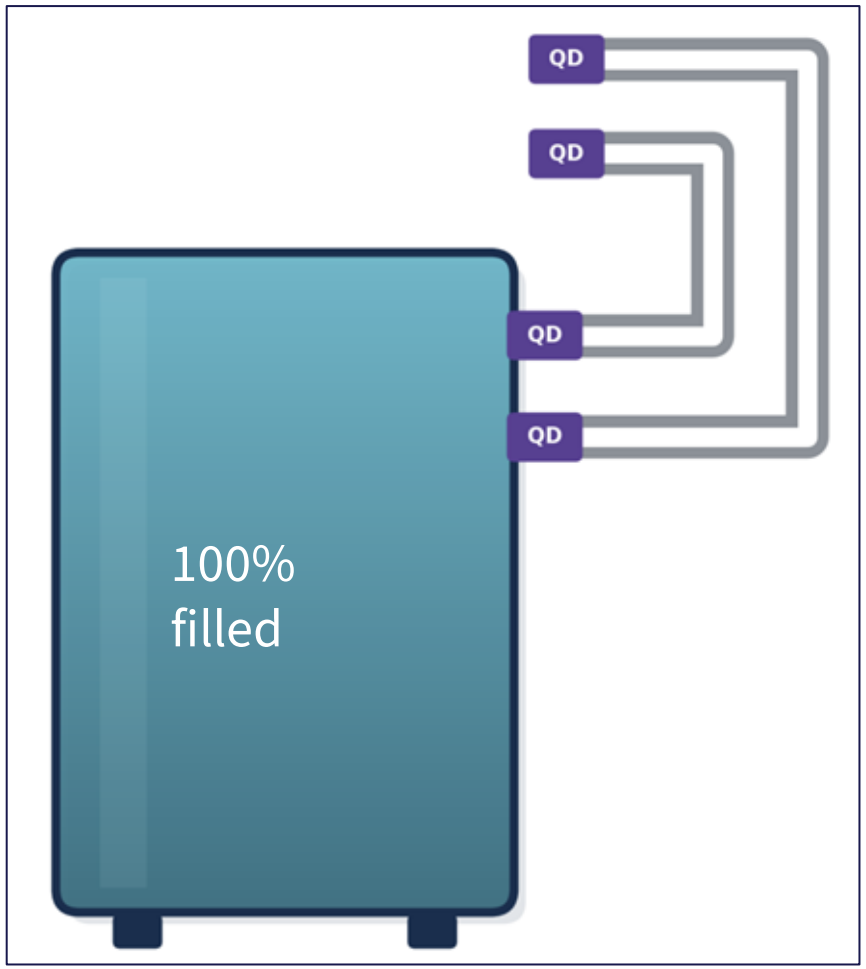
Fluid Management

Partial Wet / Partial Dry

- Passivation film persists from prior service
- Residual inhibited fluid maintains protection
- Defined dwell time and fluid inspection on return-to-service



Jumper Hose



- Jumper hose improves hardware robustness against expansion damage
- Convenient to adopt in production
- Avoids air/liquid/copper intersections

More Operational Solutions

- Nitrogen Protection
- Shelf-life control
- Pressure assisted draining
- Circulated filling

Recommendation

Empirical observations

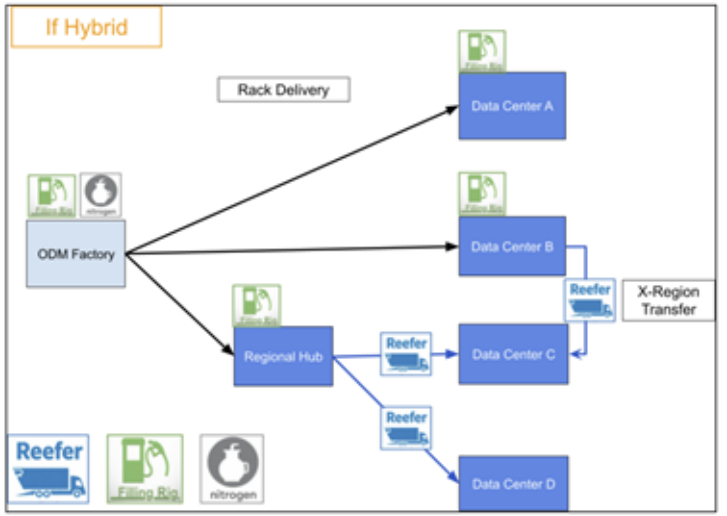
- Early Sample Racks - Partial wet
- MP Racks - Wet or Dry
- Server FRU inventory: Wet or Dry
- Server RMA: Wet or Semi Dry

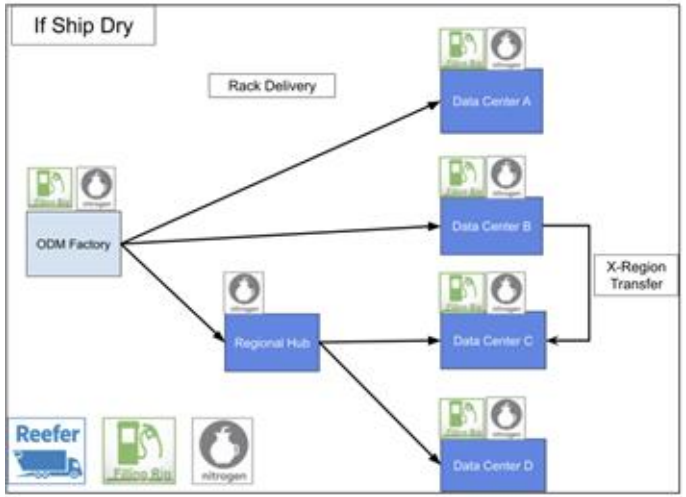
Consider logistic resource, operational resource and hardware design.

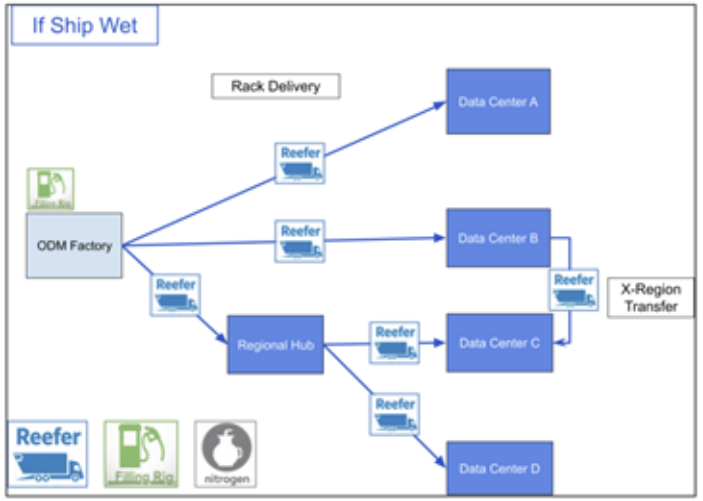
Choose, combine or adjust the mode(s) as needed.

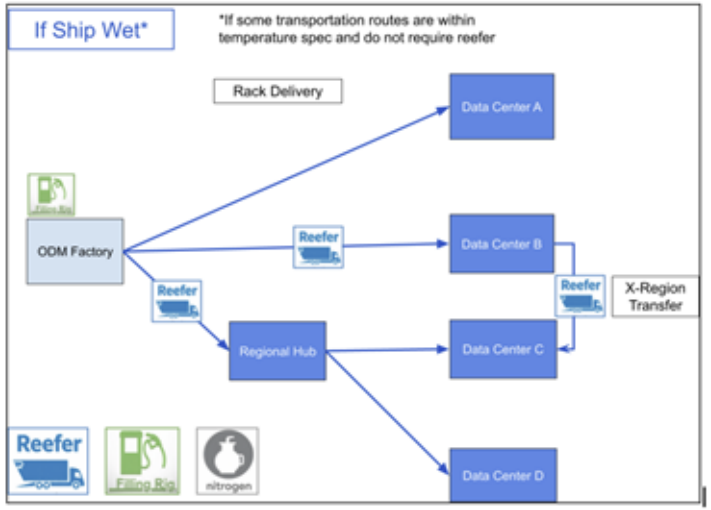
Examples

- More than one way to delivery the product
- Different modes can be combined in practice









Challenges

Lack of consensus and optimization

- Data Center Owners need to support hardware from different suppliers of different generations
- Hardware designs are not equally robust against liquid expansion
- Lack of common best practices and equipment to ensure filling/draining quality
- Lack of quantified understandings on corrosion characteristics and shelf life limits

Call to Action

- Problem to solve
  - Diverged liquid cooled hardware transportation and storage protocols
- Additional Information
  - White paper: https://www.opencompute.org/documents/considerations-for-shipping-and-storage-of-1-phase-coldplate-liquid-cooled-racks-and-servers-final-march-2026-pdf

Thank You!

